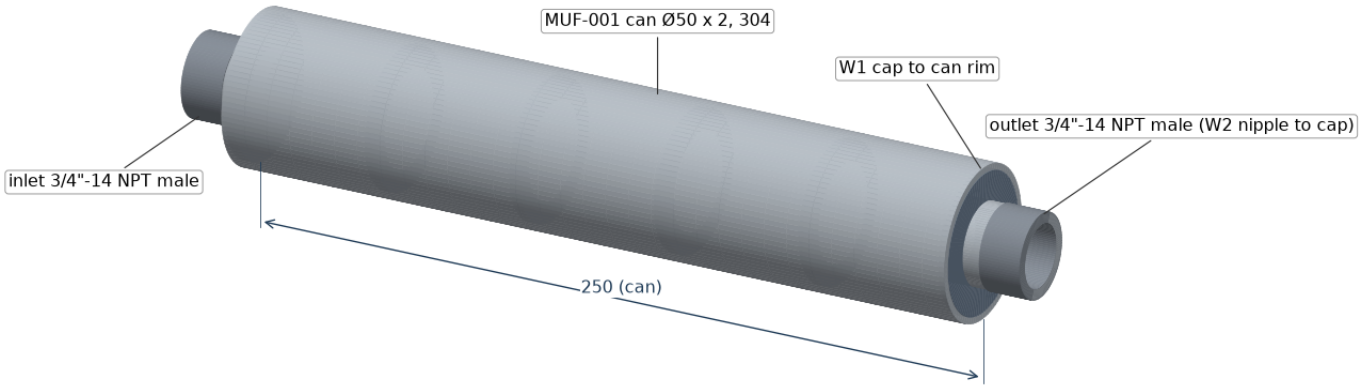
50 x 250 mm Baffle Muffler

Build Book - home TIG shop



Item	Spec
Outer can	Ø50 x 2 mm wall x 250 mm, 304
Liner	Ø32 x 2 mm wall x 244 mm, 304 tube (bought, no seam), captured between caps
Perforation	480 x Ø4 mm, 16/row, 30 rows @ 7 mm, 11.25° stagger; 28.1% open (ID flow-path basis) / 24.6% (OD); web 1.50 mm
Ports	3/4"-14 NPT male both ends: weld-in 304 TOE nipple, 6 mm thread relief
Caps / baffles	3 mm 304 discs: 2 caps + 4 baffles @ 50, 100, 150, 200 mm; OD = measured can ID - 0.2
Welding	TIG DCEN / pulse, ER308L 1/16 in, max 100 A (machine 110 A, 220 V)
Material	AISI 304 throughout (316/316L + ER316L for salt/marine)

Contents: 1 How to Build - 2 Drawings - 3 3D Views - 4 TIG Parameters - 5 BOM + Sourcing - 6 Verification Checklist + Design Checks

1. How to Build

Work top to bottom. Each step lists the exact tools. Settings for every weld are in section 4.

Step 1. Measure

- Can ID at 4 clock positions, both ends -> smallest value. Liner OD likewise -> largest.
- Re-run `muffler.py --can-id <ID> --liner-od <OD>`; discs become OD = ID - 0.2, bore = OD + 0.2.

Tools: Calipers with ID jaws (0.01 mm), laptop with muffler.py

Step 2. Cut

- Can to 250 +/-0.5 and liner to 244 +/-0.3, ends square within 0.3 (check with a square). Deburr.
- Cut the two 3/4"-14 NPT TOE nipples to 28 overall from the plain end. Deburr.
- Order/cut discs: 2x MUF-003 caps (port bore 26.9), 4x MUF-004 baffles. Cut the JIG-001 guide plate.

Tools: Bandsaw or abrasive cut-off saw, machinist square, flat file, deburr tool, marker, pipe vise for the nipples

Step 3. Jig - build the centering sleeve per `jig_build.md`.

Tools: Drill press, V-block, Ø4 mm cobalt drill, TIG/MIG for tacking block + M6 nut, M6 thumb screw, file

Step 4. Drill (liner)

- Wrap guide plate, edge flush with the inlet end, tape tight. Center punch all 480 pilot holes. Remove.
- Sleeve on liner, Ø4 pin through guide bushing into each punch mark, lock thumb screw, drill Ø4 straight through the guide. ~950 rpm for Ø4 in 304, cutting fluid, steady feed (304 work-hardens if the drill rubs); swap drills as soon as they squeal.
- Deburr OD (file/flap wheel) and ID (round file, brush). Blow out chips.

Tools: Drill press, V-block, centre punch + hammer, Ø4 mm pin, Ø4 mm cobalt jobber drills (5+), cutting fluid, round file, deburr tool

Step 5. Clean - acetone on all parts; stainless-only brushes; no grinding discs used on carbon steel.

Tools: Acetone, lint-free rags, stainless-only wire brush, nitrile gloves

Step 6. Fit

- Slide baffles onto liner at z = 50, 100, 150, 200 (centres) using spacer blocks (45.5 mm from the liner's inlet end, 47 mm between baffles). Baffles sit on undrilled bands.
- Dry-fit liner + baffles into can, caps + nipples in both ends: caps flush, spigots enter the liner bore, liner captured with no end play.

Tools: Spacer blocks 48.5 mm and 47 mm, square, calipers

Step 7. Weld W3 (baffles to liner)

- Foil-tape both liner ends, argon purge the bore 4-6 L/min.
- Check square, 4 tacks per baffle @ 90 deg, re-check square, continuous fillet on the upstream face.

Tools: TIG (DCEN/pulse), argon + purge line, foil tape, ER308L 1/16 in, stainless brush

Step 8. Weld W2 (nipples to caps, on the bench)

- Nipple through the cap bore: spigot 5 mm proud of the inner face, 6 mm plain relief outside.
- Thread a 3/4"-14 NPT pipe cap on to protect the threads; purge through the nipple.
- 4 tacks, check square, continuous fillet on the outside face, on the plain section only.

Tools: TIG, argon purge, 3/4"-14 NPT pipe cap (thread guard), square, ER308L 1/16 in

Step 9. Fit can - slide the liner/baffle core into the can. No packing: absorption is by the baffles only.

Tools: Rubber mallet, gloves

Step 10. Cap

- Press caps in flush both ends; spigots enter the liner bore. Purge through a nipple (the perforations carry argon into the chambers).
- 4 tacks each, then W1 (cap to can rim), alternating ends to limit distortion.

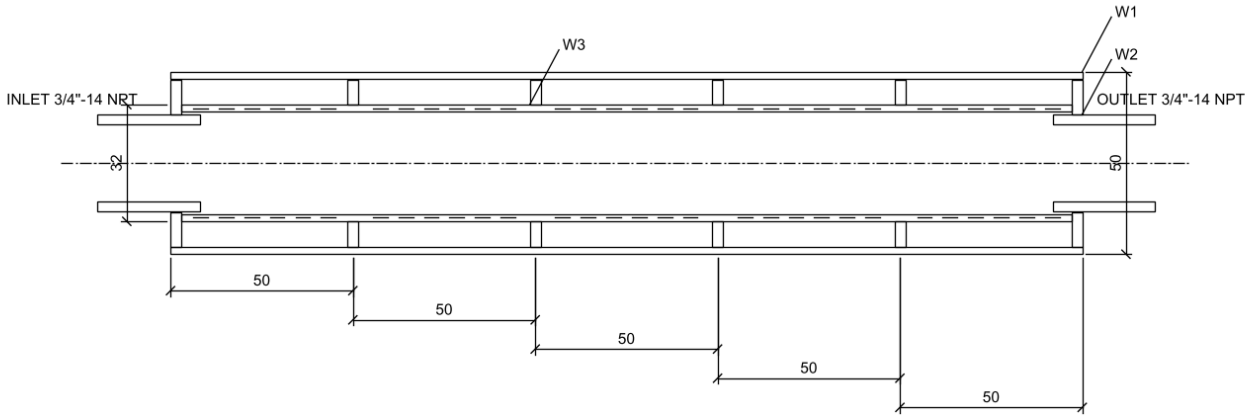
Tools: TIG, argon purge, foil tape, clamps, ER308L 1/16 in

Step 11. Finish - pickle/passivate welds; chase threads with a mating fitting; run `verification_checklist.md`.

Tools: Pickling paste (acid: gloves + eye protection), mating 3/4"-14 NPT fitting, pipe cap, plug with Schrader valve, soapy water, brush

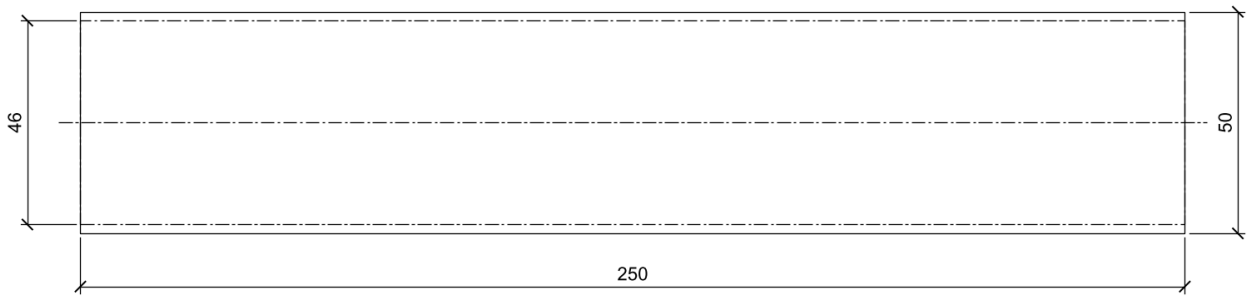
2. Drawings

Full-size DXF files are in the package for laser/print; these are previews. Units: mm.



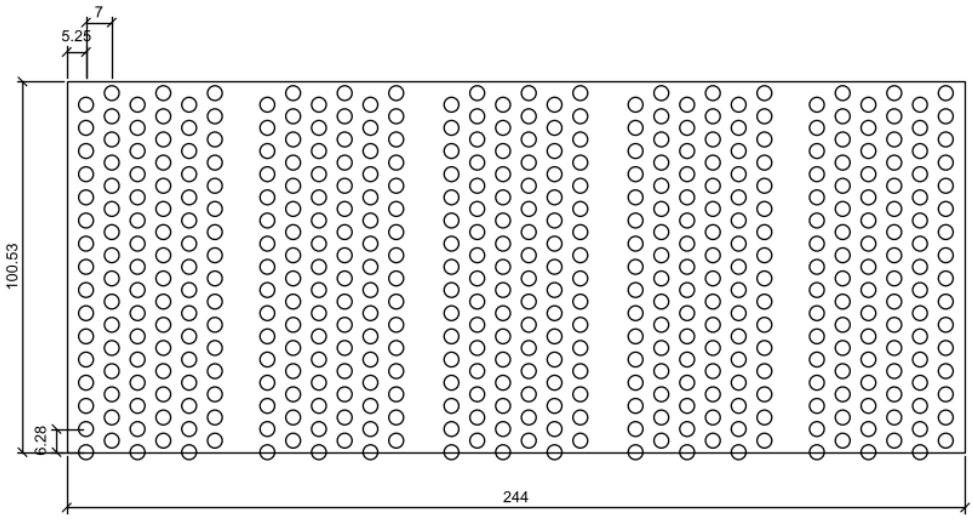
MUF-000 ASSEMBLY HALF-SECTION
MATL: AISI 304 (316 for corrosive/salt service)
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER/DETAIL = do not cut
W1 cap-can rim | W2 nipple-cap | W3 baffle-liner (see weld map)
Liner 244 long, captured between caps; nipples spigot 5 into liner bore. No packing.

MUF-000 Assembly half-section: can, captured liner, caps with 3/4"-14 NPT nipples, 4 baffles; weld callouts W1-W3.



MUF-001 OUTER CAN
MATL: AISI 304 (316 for corrosive/salt service)
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER/DETAIL = do not cut
TUBE 50 OD x 2 WALL x 250 LG, ENDS SQUARE WITHIN 0.3, DEBURR
MEASURE ID AT 4 CLOCK POSITIONS BOTH ENDS; USE SMALLEST FOR DISC OD

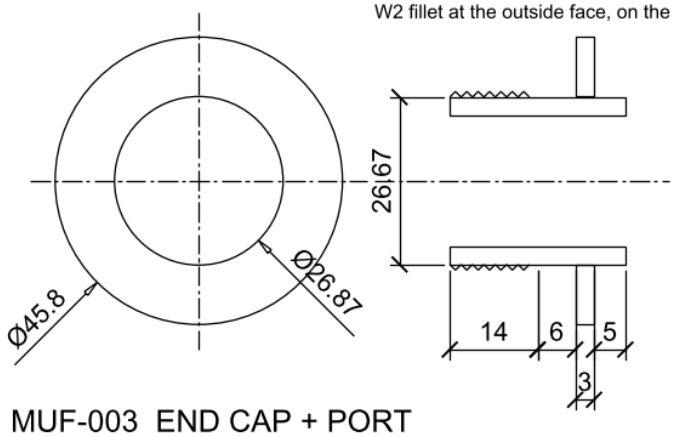
MUF-001 Outer can: 50 OD x 2 wall x 250 tube, ends square, measure ID x4.



MUF-002 PERFORATED LINER - WRAP LAYOUT (OD)
MATL: AISI 304 (316 for corrosive/salt service)
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER/DETAIL = do not cut
PURCHASED TUBE 32 OD x 2 WALL x 244 LG, 304 (seamless preferred). X FROM INLET END
480 HOLES DIA 4 THRU RADIAL, 30 ROWS, 16/ROW @ 22.5 deg
ALTERNATE ROWS ROTATED 11.25 deg (HALF PITCH STAGGER). Y = arc on OD from datum line
OPEN AREA 28.1% (ID BASIS) / 24.6% (OD). DEBURR ID AND OD. DRILL WITH JIG (drill_jig.dxf)

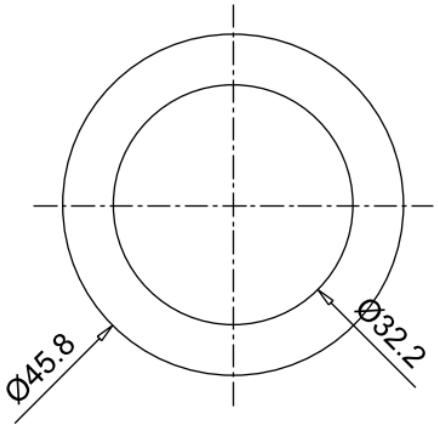
MUF-002 Liner wrap layout (OD), 244 long: 480 x Ø4 holes, 16/row, 30 rows, 11.25° stagger.

SECTION: 3/4"-14 NPT TOE NIPPLE IN CAP
thread 14 (L2 13.86) | relief 6 | cap 3 | spigot 5 into liner
W2 fillet at the outside face, on the plain relief - never on threads



MUF-003 END CAP + PORT
MATL: AISI 304 (316 for corrosive/salt service)
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER/DETAIL = do not cut
THK 3 sheet, QTY 2. OD = MEASURED CAN ID - 0.2
PORT BORE 26.9 for 3/4"-14 NPT nipple (OD 26.67), 304 sch40, TOE, cut to 28
Thread relief: no thread within the weld zone; weld on the plain end only

MUF-003 End cap (x2): 3 mm disc, port bore 26.9; section shows the 3/4"-14 NPT nipple, thread relief and spigot.



MUF-004 BAFFLE

MATL: AISI 304 (316 for corrosive/salt service)

UNITS: mm TOL: +/-0.2 unless noted

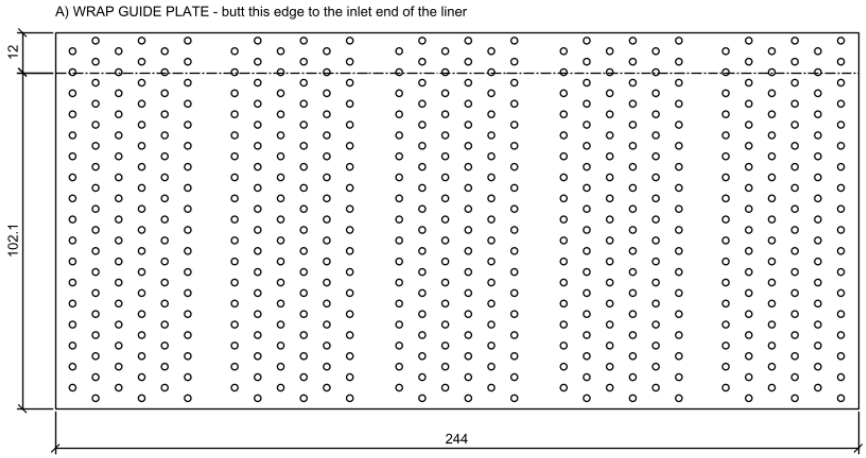
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER/DETAIL = do not cut

THK 3 sheet, QTY 4

OD = MEASURED CAN ID - 0.2; BORE = MEASURED LINER OD + 0.2

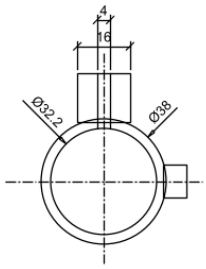
Stations z = 50, 100, 150, 200

MUF-004 Baffle (x4): same 3 mm blank as the cap, stations at 50/100/150/200.



0.5 mm shim (Al/steel) or 1:1 print glued to card. Pilot holes DIA 2.

Tab overlaps start; tab holes coincide when wrapped. Circumference at shim mid = 102.10



B) CENTERING SLEEVE (end view)

Sleeve: 38 OD tube, bore 32.2, 40 long

Bushing block 16x15x40 steel, guide hole DIA 4 RADIAL through axis; M6 nut + thumb screw on side to lock

JIG-001 DRILL JIG

MATL: shim: 0.5 Al/steel; sleeve + block: mild steel

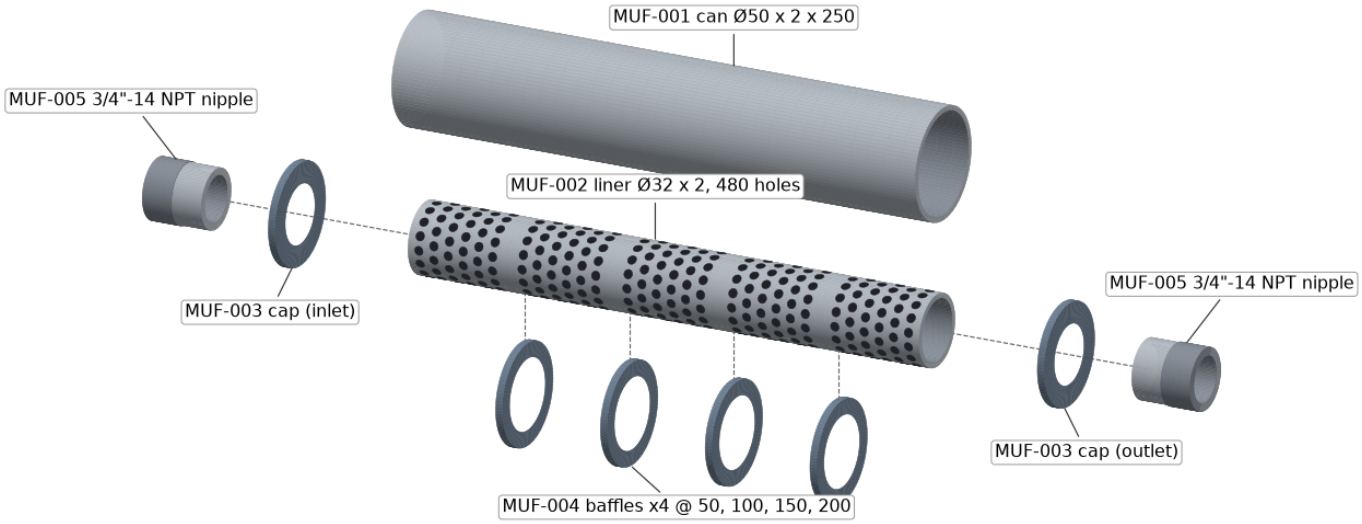
UNITS: mm TOL: +/-0.2 unless noted

LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER/DETAIL = do not cut

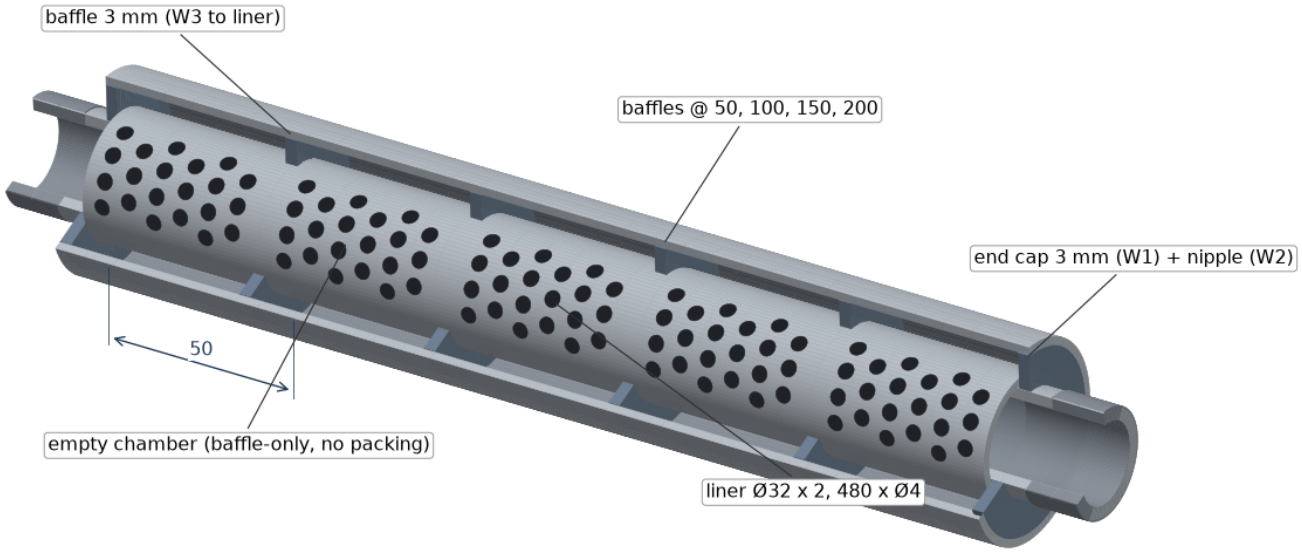
480 pattern positions = liner pattern (480)

JIG-001 Drill jig: 0.5 mm wrap-around shim (480 marks) + centring sleeve with Ø4 guide.

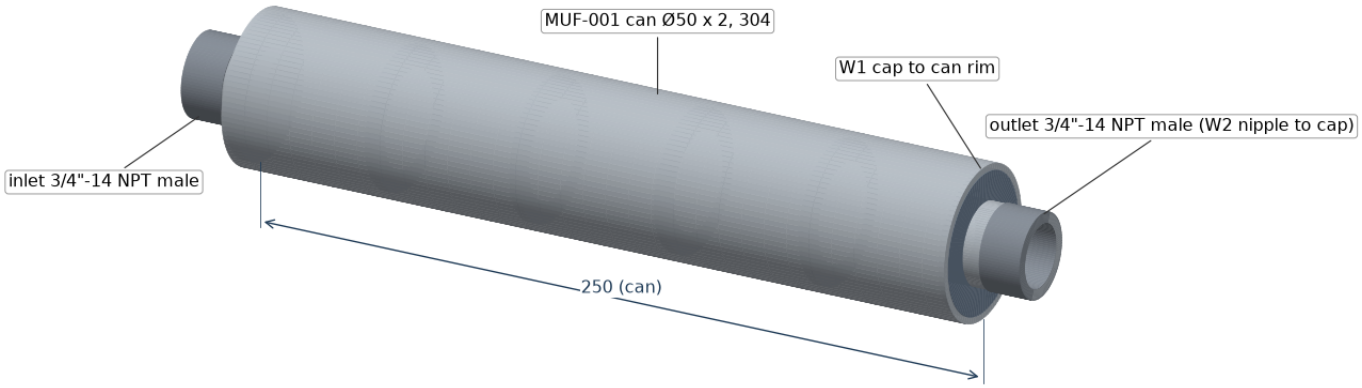
3. 3D Views



Exploded: liner with 480 holes in the middle, baffles below, can above, caps and port nipples on the ends. Dashed lines show where each part goes.



Cut-away: quarter removed to show the 4 baffles, the empty chambers between them and the perforated liner.



Assembled: caps flush with the can ends; 3/4"-14 NPT male nipples are the inlet and outlet.

4. TIG Parameters

Shop machine: 220 V input, 110 A max, AC/DC/pulse, back-purge available.

ID	Joint	DC (DCEN)	Pulse (recommended)	Argon	Back-purge	Filler
W1	End cap OD to can rim	65-85 A	base 40-60 A, peak 100 A, 30-50 Hz	8-10 L/min (gas lens 10-12)	4-6 L/min	ER308L 1/16 in
W2	Port nipple (3/4"-14 NPT) to end cap	80-100 A	base 60-80 A, peak 100 A, 30-50 Hz	8-10 L/min (gas lens 10-12)	4-6 L/min	ER308L 1/16 in
W3	Baffle ID to liner OD	80-100 A	base 60-80 A, peak 100 A, 30-50 Hz	8-10 L/min (gas lens 10-12)	4-6 L/min	ER308L 1/16 in

- **DCEN for all joints.** AC is for aluminium; don't use it on stainless.
- **Pulse is recommended** on 2 mm 304: lower heat input means less burn-through at hole webs and less sugaring.
- W2 (2.9 mm nipple wall to 3 mm cap) and W3 (2 mm liner to 3 mm baffle) are both tube-to-disc fillets, so they
- Peak current never exceeds 100 A, which leaves headroom on the 110 A machine.
- Gas: Ar 8-10 L/min (10-12 with a gas lens); postflow 8-10 s. 2% lanthanated 1/16 in tungsten, #7 cup.
- **Back-purge 4-6 L/min on every hollow joint** (W3 on the liner, W2 through the nipple, W1 through a nipple
- Tack first (4 per circular joint), then continuous. No preheat for TIG on 304. Interpass < 150 C.
- No W4: the liner is bought tube, so there is no shop seam. With welded tube the factory seam may be drilled through.

Weld map

ID	Joint	Qty	Type / extent	Size
W1	End cap OD to can rim	2	Flush corner (butt-style), full 360, outside	2 (= can wall)
W2	Port nipple (3/4"-14 NPT) to end cap	2	Fillet, full 360, outside face	2 leg
W3	Baffle ID to liner OD	4	Fillet, full 360, upstream face	2 leg

No W4: the liner is bought tube, so there is no shop seam.

5. BOM + Sourcing

Part	Name	Material	Qty	Notes	Source
MUF-000	Muffler assembly	-	1	Weldment; MUF-000 DXF, weld map, build book	-
MUF-001	Outer can	304 tube 50 OD x 2 wall	1	Cut 250 +/-0.5, square ends, deburr. Measure ID x4.	Online Metals 50x2 304, 60 in length (MTR available) or Amazon 50x2x400 mm (cut to 250)
MUF-002	Perforated liner	304 tube 32 OD x 2 wall (seamless pref.)	1	Cut 244 +/-0.3; 480x DIA 4 radial per drilling spec, using JIG-001; deburr ID+OD	AliExpress 32x2 304 tube, 300 mm (cut to 250) or uxcell 32x2 on Amazon. Fallback rolled 2 mm sheet = 7 holes/row + W4 seam variant (not generated here)
MUF-003	End cap	304, 3 sheet	2	Laser: OD 45.8, port bore 26.9 for the nipple	3 mm 304 plate; laser-cut service (SendCutSend, OSH Cut, local shop) from MUF-003/004 DXF
MUF-004	Baffle	304, 3 sheet	4	Laser: OD 45.8, bore 32.2	3 mm 304 plate; laser-cut service (SendCutSend, OSH Cut, local shop) from MUF-003/004 DXF
MUF-005	Port nipple 3/4"-14 NPT	304 sch40 pipe nipple, thread one end (TOE), 2 in long	2	Cut plain end to 28 overall (14 thread + 6 relief + 3 cap + 5 spigot); deburr	McMaster-Carr / plumbing or industrial supply (304 TOE nipple)
MUF-006	TIG filler rod	ER308L 1/16 in (1.6 mm); ER316L if 316	1	~0.25 kg tube	Welding supply
JIG-001	Drill jig	0.5 mm shim + 38 OD tube + steel block + M6 nut	1	See jig_build.md	Shim stock + scrap

• **Liner** (32 x 2 304 tube): AliExpress 32x2 304 tube, 300 mm (cut to 250) or uxcell 32x2 on Amazon. Fallback rolled 2 mm sheet = 7 holes/row + W4 seam variant (not generated here).

• **Can** (50 x 2 304 tube): Online Metals 50x2 304, 60 in length (MTR available) or Amazon 50x2x400 mm (cut to 250).

• **Discs** (caps + baffles): 3 mm 304 plate; laser-cut service (SendCutSend, OSH Cut, local shop) from MUF-003/004 DXF. Order **after** measuring the tube (step 1).

• US imperial near-equivalents if metric is short: 2 in x 0.083 in (can), 1-1/4 in x 0.083 in = 31.75 x 2.11 (liner);

• 316/316L if the vehicle sees road salt or marine use; then use ER316L filler.

• Filler ER308L 1/16 in (ER316L for 316), 2% lanthanated tungsten, argon.

6. Verification Checklist

Part: 50 x 250 baffle muffler. Builder: _____ Date: _____

Liner (before assembly)

[ ] Hole count = **480** (30 rows x 16). Count by row, tick each row off.

[ ] Hole Ø4 +0.3/-0, round (caliper in two directions).

[ ] Web check: thinnest web between holes >= 1.4 mm (check on the ID; design 1.50). Axial web between rows >= 2 mm (design 3.00).

[ ] No hole within 3 mm of baffle/cap positions (z = 50, 100, 150, 200 and both ends).

[ ] ID and OD deburred, no chips inside.

Fit

[ ] Can ID measured x4 each end: smallest _____ ; disc OD = ID - 0.2 = _____ (design 45.8).

[ ] Cap fit: cap enters can by hand pressure, gap <= 0.3 all round, sits flush with the can end.

[ ] Baffles slide on the liner without forcing; liner 244 mm, captured between caps with no end play.

Port (3/4"-14 NPT)

[ ] Nipple thread 14 mm long, 6 mm plain relief outside the cap, spigot 5 mm into the liner.

[ ] Thread gauge / mating fitting runs on by hand after welding (no weld spatter or heat damage on threads).

[ ] Baffle spacing: centres at 50, 100, 150, 200 +/-1 mm, square to axis within 0.5 mm at OD.

Welds

[ ] W3 x4, W2 x2, W1 x2 complete, full 360, no cracks/porosity/undercut; colour straw-blue, not grey.

[ ] Soapy-water leak test: 3/4"-14 NPT pipe cap on the outlet, plug with Schrader valve in the inlet, **0.2-0.3 bar (3-4 psi) max**. Brush soap on every W1 and W2 joint: no bubbles in 30 s. (W3 is internal; leaks between chambers don't matter.)

Final

[ ] Rattle check: shake hard by hand; nothing loose.

[ ] Sound check: engine at idle and 3/4 throttle, phone dB meter at 0.5 m, 45 deg from the outlet; record without / with muffler: _____ / _____ dB(A).

[ ] Welds passivated, no rust bloom after 24 h.

Design checks (computed by muffler.py)

Check	Value	Result
Hole count	480 = 30 rows x 16	PASS
Web, within row @ ID (worst)	1.50 mm >= 1.4	PASS
Web, within row @ OD	2.28 mm	PASS
Web, between rows (axial)	3.00 mm >= wall 2	PASS
Web, diagonal (11.25 deg offset) @ ID	3.52 mm >= 1.4	PASS
Open area, whole liner	28.1% on ID (28 mm) flow-path basis, target 27-33%; 24.6% on OD (32 mm) basis	PASS
Hole edge to cap/baffle face	3.25 mm >= 3.0	PASS
Disc OD in can ID	45.80 < 46.00	PASS
Baffle bore over liner OD	32.20 > 32.00	PASS
Port 3/4"-14 NPT: nipple spigots into liner	nipple OD 26.67 < liner ID 28; cap bore 26.9	PASS
Port thread engagement + weld relief	thread 14 >= L2 13.86; relief 6 >= 6	PASS
Baffle count 3-5	4	PASS
Max TIG amps (DC / pulse peak) <= welder	100 A <= 110 A	PASS
Guide plate pattern = liner pattern	480 = 480	PASS